Apply filters to SQL queries

Project description

Through SQL, I examined a hypothetical organization's data to investigate security issues and helped to keep their system secure. I used the `log_in_attempts` and `employees` tables.

The `log_in_attempts` table has the following columns:

```
+----------+----------+------------+------------+---------+-----------------+---------+
| event_id | username | login_date | login_time | country | ip_address      | success |
+----------+----------+------------+------------+---------+-----------------+---------+
```

The `employees` table has the following columns:

```
+-------------+---------------+----------+-----------------------+-------------+
| employee_id | device_id     | username | department            | office      |
+-------------+---------------+----------+-----------------------+-------------+
```

Retrieve after hours failed login attempts

In this scenario, I need to investigate a potential security breach that occurred after business hours. I will need to filter through the `log_in_attempts` table to identify all failed login attempts that occurred after 18:00. I need to use the following query to achieve this:

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_time > '18:00' and success = '0';
```

Success having a value of 0 means that the login attempt was unsuccessful.

Retrieve login attempts on specific dates

To investigate an event in which suspicious login attempts were made on specific dates, such as `2022-05-08` or `2022-05-09`, you would use this query:

```
MariaDB [organization]> SELECT*
    -> FROM log_in_attempts
    -> WHERE login_date = '2022-05-08' OR login_date = '2022-05-09';
```

Retrieve login attempts outside of Mexico

If this organization were to reside in Mexico, and there's been suspicious activity outside of Mexico, you would use this query to find foreign login attempts:

```
MariaDB [organization]> SELECT*
    -> FROM log_in_attempts
    -> WHERE NOT country LIKE 'MEX%';
```

"Mex%" must be used instead of "Mexico" as the country is sometimes regarded as "Mex" or "Mexico", so using `LIKE 'MEX%'` includes all countries that start with "Mex".

Retrieve employees in Marketing

If your security team were to want to identify all employees in the 'Marketing' department that work in the "East" building, you would build this query:

```
MariaDB [organization]> SELECT*
    -> FROM employees
    -> WHERE department = 'Marketing' AND office LIKE 'East%';
```

Again, both `LIKE` and `%` are used to specify that we want anything including `East`, such as the `East-170` or `East-267` offices.

Retrieve employees in Finance or Sales

In a scenario in which you would need to identify all employees in either the Finance or Sales departments, this query should be used:

```
MariaDB [organization]> SELECT*
    -> FROM employees
    -> WHERE department = 'Finance' or 'Sales';
```

Retrieve all employees not in IT

Finally, to retrieve all employees not in a single department, like IT, you would build this query:

```
MariaDB [organization]> SELECT*
    -> FROM employees
    -> WHERE NOT department = 'Information Technology';
```

Summary

I filtered records to surface after-hours failed logins, activity on specific dates, and employee access by department and location. This lab demonstrates how SQL supports threat investigation and access review.